

Multiscale Neural Organization and Conscious Closure in the Balance Field Framework

An ontological synthesis with conditional mathematics and testable EEG hypotheses

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Abstract

The Balance–Field Framework (BFG) proposes that differentiated processes form persistent units through neutral mediation and recursive closure. This paper integrates six BFG manuscripts into a standalone account of multiscale neural organization, intentionality and disturbances of consciousness. It separates an ontological interpretation of inward and outward organization from a structural account of distinguishability, an operational neural hypothesis and conditional mathematical results. A hierarchy of local order states prevents global balance from concealing differences between organizational levels. A six-dimensional linear-response specialization represents content and self-related roles at three coupled levels through a positive neutral operator and conjugate recursive dynamics. Its spectrum is derived exactly. Simulations show that identical spectral radius and common-mode retention coexist with absent cross-level propagation or a reduction of contrast retention by more than five orders of magnitude. A three-dimensional stability surface and perturbation trajectories illustrate these distinctions. An exact direct-coupling clone reproduces all baseline outputs, establishing that the chosen neutral factorization is not physically identified by those outputs. The resulting EEG program therefore requires independent role calibration, interventions with justified causal targets, matched model comparisons and prospective transfer. Existing studies of perturbational complexity, cognitive motor dissociation, perceptual coordination and speech-related coupling motivate separate empirical tests. The contribution is a coherent BFG research architecture with transparent assumptions and reproducible consequences. It does not establish that recursive stability is sufficient for consciousness, that a particular neural mediator has been discovered, or that a clinical biomarker has been validated.

Keywords: Balance–Field Framework; multiscale organization; neutral mediation; recursive closure; intentionality; EEG; TMS–EEG; identifiability; consciousness; mathematical modelling.

1 Introduction

A theory connecting consciousness and neural activity must explain the organization, persistence and physical realization of its object. For intentionality—the directedness of experience toward an object, situation or goal—temporal structure alone does not explain content or subjective reference. A philosophical account of unity likewise requires a further bridge to a measurable neural mechanism.

Fuchs describes disturbances of intentionality in schizophrenia through a disruption of the integration of past, present and anticipated continuation. His account connects phenomenological analysis with neuropsychological findings; it is not simply a description of EEG timing [7]. The present paper treats this account as a useful constraint. A proposed neural realization should explain how content remains connected and context-sensitive through change, while preserving the distinction between a functional mechanism and the experience whose organization it may support.

The BFG corpus contains complementary approaches to this problem. The direction-error manuscript interprets physical appearance and inward experience as aspects of deeper recursive organization [1]. The joint Wende–Pereira manuscript examines conditions of distinguishability and conscious unity [2]. The order-indexed interface paper assigns local order and mediation to a hierarchy of closures [3]. The neural recursive closure manuscript proposes world-related and self-related neural roles [4]. The qualia-closure manuscript separates an operational witness from the quale to which it may be related [5]. The integrated reclosure preprint strengthens the distinction between conditional mathematical construction, physical identification and irreducible emergence [6].

Our aim is to assemble these contributions without converting one paper’s interpretive premise into another paper’s theorem. We develop an explicitly declared neural-response specialization, derive its consequences, demonstrate a failure of scalar stability as a complete organizational description, and specify the evidence needed to move from an effective operator to a physically identified mechanism. Standard linear algebra is used to analyze the model. The neural assignments and parameter choices are additional modelling assumptions; they are not presented as deductions from historical Level 0.

Disturbances of consciousness, psychiatric disorders and psychodynamic processes require separate outcome measures. The proposed variables do not define a universal clinical label.

2 Source synthesis and contribution

Table 1. BFG source roles and the boundaries retained in this synthesis.

Source	Retained contribution	Boundary
[1] Direction error	Inward and outward aspects of recursive organization	Ontological interpretation
[2] Conditions of possibility	Distinguishability, identity retention and self-indexing	Conditional explanatory roles
[3] Order-indexed interface	Local order fields and bidirectional transport	Carrier maps remain to be specified
[4] Neural recursive closure	Content, self, mediation and EEG hypotheses	No clinical validation
[5] Qualia closure	Operational witness separated from experience	Decoding is not phenomenal identity
[6] Integrated reclosure	Source assumptions, interventions and exact clones	No universal natural emergence result

The new contribution is the explicit connection between level-resolved organization, common-versus-contrast response modes and an exact identifiability control. The six-state construction below is introduced

here as an analytical illustration. It is not a replacement for the source papers' full dynamics, and it does not establish the compatibility of all historical BFG variants. In particular, the real tangent postulate and successor choices discussed in the latest reclosure manuscript [6] do not independently derive content, selfhood, the EEG observation map or the six-state coupling law used here.

Three evidential levels are used throughout. A structural proposal states what roles BFG assigns to organization. A conditional result states what follows from explicitly supplied mathematical assumptions. An empirical claim requires observations independent of the model's construction. Our numerical work occupies the second level. External literature supplies established examples, methods and competing explanations; citation of that literature does not independently validate BFG.

3 Ontological interpretation and conscious organization

3.1 The inward and outward interpretation

The strongest ontological reading of [1] treats outwardly observable structure and inward experience as aspects of one underlying recursive organization. This changes the proposed explanatory starting point. It does not eliminate the need to justify the relation between organization and experience. The explanatory gap identified in the consciousness literature remains a challenge whenever a formal description is claimed to entail subjective presence [8]. Naming a dynamical regime "inward" cannot provide that entailment by itself.

The Wende–Pereira manuscript adopts a more restrained role-level formulation [2]. Neutral admissibility is a proposed enabling condition for distinguishability. It is not identified there with a measurable force, field, substance or primitive observer. Its necessity argument depends on requiring an explanation of distinguishability rather than taking distinguishability as primitive. A rival may reject that explanatory demand or satisfy it differently. Thus the ontological interpretation of [1] and the dependency argument of [2] should not be treated as equivalent conclusions.

3.2 A dependency architecture

$$N_0 \rightarrow \Delta_{N_0} \rightarrow I \rightarrow C_{candidate}. \quad (1)$$

Equation (1) summarizes the proposed dependency architecture: neutral admissibility, admissible difference, stabilized information and candidate conscious closure. The arrows express explanatory dependence in the BFG interpretation. They are not a measured chronology of brain development or a derived physical evolution equation. The final step adds requirements concerning unity, identity through change and self-related organization [2]. None of these requirements is inferred solely from the existence of a stable fixed point.

This distinction matters for circularity. If a model starts by assuming a sentient subject, it may investigate how that subject organizes content; it cannot then claim to have derived the existence of sentience without further premises. Here, world-related and self-related variables are proposed functional roles. Their presence in a model is not taken as proof of a phenomenal subject.

3.3 Experience and its operational witness

$$W_x = Inv_G^0(C_x), W_x \neq Q_x \text{ by definition.} \quad (2)$$

Following [5], C_x denotes a candidate content-closure organization, Q_x its hypothesized qualitative identity and W_x a fixed operational witness calculated from the organization. The inequality in Equation (2) records a distinction of explanatory role, not a numerical test. A reproducible indicator may support discrimination of conditions without being identical to what red looks like, what pain feels like or what it means to intend

an action. Reports, task performance and physiological responses should constrain the model together, with their dependencies made explicit.

For intentionality, the proposal is therefore specific but incomplete: a content-bearing process should remain differentiated, be embedded in a context-sensitive relation to action or perception, and maintain a reproducible relation to self-related organization. The additional claim that this realizes phenomenal aboutness is a bridge hypothesis. The finite-dimensional construction in Sections 6–8 tests only dynamical prerequisites of that hypothesis.

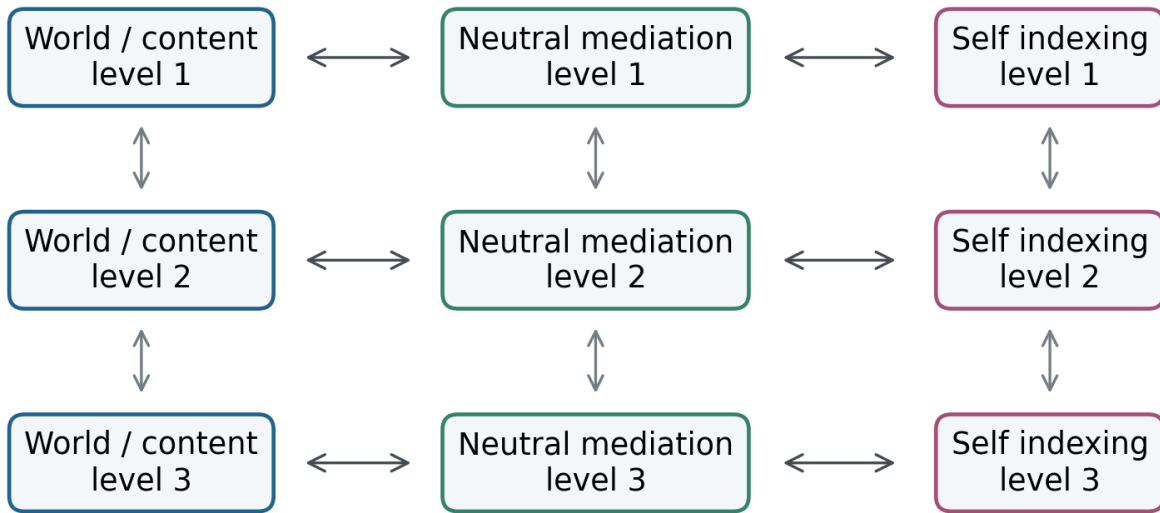
4 Multiscale closure architecture

4.1 Local order and mediation

$$T_n = (P_{1,n}, P_{2,n}, N_{rel,n}, \hat{N}_n[\varphi_n], K_n, C_n), \varphi_n > 0. \quad (3)$$

At level n , the structural object contains two distinguishable poles, a mediating relation, its proposed operator realization, a configuration K_n and a closure C_n . The neutral relation, neutral operator and local order coordinate are different objects. They must not be identified merely because they share the word neutral. Figure 1 illustrates the role structure. A level is an organizational scale specified by its variables and transformation rules; it is not automatically an EEG frequency band, anatomical layer or successive time window.

Role architecture and cross-level return



Hypothesized roles; levels are not identified with cortical layers or EEG frequency bands.

Figure 1. BFG role architecture across three illustrative levels. Within-level mediation and cross-level return are proposed functional relationships. The diagram makes no anatomical localization claim.

$$\Phi_{src,n} = \frac{E_{exp,n} I_n}{E_{bind,n} I_{0,n}}, \varphi_n v_n(\varphi_n) = \Phi_{src,n}. \quad (4)$$

Equation (4) retains the source balance ratio and the interface-resolved order equation from [3]. The reference information scale must be positive, and the expansion and binding quantities must have compatible dimensions. A physiological application must independently specify these quantities; neither metabolic power nor signal variance can simply be relabelled as one of them. The response function v_n and

the operator from which it is obtained require a constitutive model. Their functional forms are not selected by the ratio alone.

4.2 Global summaries and hidden local differences

$$\Phi = (\varphi_1, \dots, \varphi_L), \Phi_G = \exp\left(\sum_n w_n \log \varphi_n\right), w_n > 0, \sum_n w_n = 1. \quad (5)$$

$$V_\Phi = \sum_n w_n (\log \varphi_n - \log \Phi_G)^2. \quad (6)$$

The positive weights in Equation (5) are normalized. $V\Phi$ is nonnegative and vanishes exactly when all local order states with positive weight coincide. It therefore detects a difference suppressed by the geometric mean [3]. This is an organizational distinction before any clinical interpretation is attached.

Proposition 1. A global balance summary does not determine the distribution of local order. **Proof.** With three equal weights, both (1, 1, 1) and (1/2, 1, 2) have geometric mean 1. Their dispersions are respectively 0 and $2(\log 2)^2/3$, approximately 0.320302. Consequently, equality of the global scalar does not imply equality of the local organization. This counterexample establishes loss of information under aggregation; it does not by itself establish that either profile is healthy or pathological.

Equal geometric mean of 1 does not determine local order

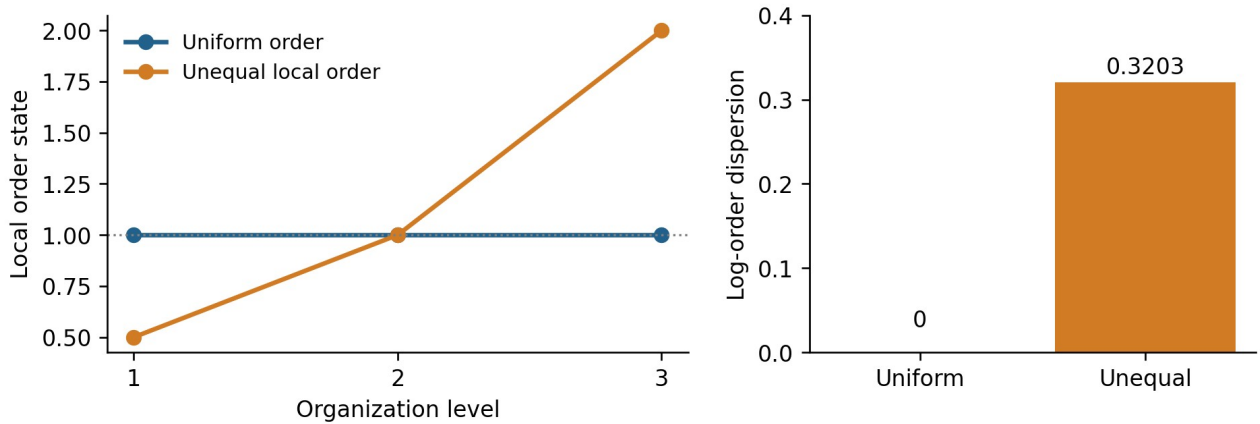


Figure 2. Two local-order profiles with the same geometric mean. The unequal profile has log-order dispersion 0.320302. These are exact mathematical examples, not measurements of brain balance.

4.3 Cross-level transport and return

$$d_{W_n}(D_{n+1,n} \iota_{n+1} U_{n,n+1}(z), z) \leq \delta_n, z \in W_n. \quad (7)$$

In Equation (7), U maps a declared witness from a lower level upward, ι embeds it in the higher representation, and D returns a compatible state. The comparison takes place in a specified witness space with a declared distance and tolerance. This follows the selective transport logic of [3]. It requires neither a return of every microscopic datum nor exact reversal of a dissipative biological process. A coarse-graining rule must specify what is retained and what can be discarded.

Temporal multiscale measures are useful but answer a different question. Multiscale entropy quantifies irregularity after temporal coarse-graining [11]; it does not, by itself, identify a hierarchy of physical causes. A BFG neural application needs spatial or population-level variables, cross-level maps and observation models in addition to any temporal complexity measure. Established dynamic causal models already provide explicit mechanisms linking hidden neural states and measured responses [9], and adiabatic dynamic causal modelling couples faster neuronal fluctuations to slower parameter changes [10]. BFG

therefore needs a distinguishable contribution rather than a claim that multiscale mechanistic modelling is absent elsewhere.

5 Neural roles and measurable predictions

$$P_{1,n} = W_n^{content}, P_{2,n} = S_n^{self}, N_n = N_n^{mediation}. \quad (8)$$

The assignment in Equation (8) specializes the neural manuscript [4]. W represents stimulus-, world- or content-related organization; S represents self-related indexing such as attribution of an action or maintenance of an internal reference; N represents the proposed mediation. These are distributed functional roles. A region can contribute to more than one role, and a role can extend over several regions. Their empirical distinction must be established rather than assumed from anatomical naming.

$$x_{t+1} = A(\theta, u_t) x_t + B u_t + \xi_t, y_t = H x_t + \epsilon_t. \quad (9)$$

Equation (9) is an operational template: hidden states evolve under input, while sensors measure a mixture of those states with noise. An EEG implementation needs a source-to-sensor or effective observation matrix H, a justified input map B and a declared noise model. The polarity roles cannot be read directly from electrode names. The toy model below sets B to the identity and omits noise for exact response calculations; it does not implement an EEG forward model.

Table 2. Proposed operational questions and major interpretive alternatives.

Role or property	Candidate test	Alternative to control
Content differentiation	Decode task content across held-out conditions	Stimulus energy or task difficulty
Self-related indexing	Self-generated versus externally replayed events	Attention, motor preparation or report
Mediation and return	Input-conditioned directed response and selective perturbation	Common drive or matched direct coupling
Cross-level organization	Source-level propagation under declared aggregation maps	Sensor mixing or arbitrary parcellation
Bounded persistence	Response decay and recovery after perturbation	Linear filtering or passive storage

Functional connectivity is not sufficient evidence of the proposed causal mediation. Volume conduction and common-source mixing can create misleading relations between EEG channels. Imaginary coherency and weighted phase-lag approaches address specific aspects of these problems [19,20], but neither uniquely identifies a BFG mediator. Any directed model also depends on the observation model, omitted causes and the conditions under which interventions can be interpreted.

$$R_C = M R M^T, M = N^{1/2}, N \geq 0. \quad (10)$$

Equation (10) records the conjugate recursive operator used in the BFG neural and interface papers [3,4]. Positive semidefinite N admits a square root. This construction is mathematically well defined once R and N are given. A physiological inference still needs evidence for their separate roles. The symbol N_0 in Section 3 denotes a pre-informational explanatory condition; it is not the same object as the finite-dimensional matrix N used here.

6 A finite linear response specialization

6.1 Declared modelling assumptions

We introduce four additional assumptions. A1: three organizational levels each have two real response coordinates. A2: within-level coupling is reciprocal and cross-level coupling follows a three-node path. A3: a positive, time-independent gain at each level scales both role coordinates. A4: we study linear deviations around a maintained configuration over a finite observation horizon. These assumptions are selected to make the consequences transparent. They are not inferred from Level 0 and do not identify neural cell types, firing rates or frequency bands.

$$x_t = (w_{1,t}, s_{1,t}, w_{2,t}, s_{2,t}, w_{3,t}, s_{3,t})^T \in \mathbb{R}^6. \quad (11)$$

$$L_3 = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}, J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (12)$$

$$R = a I_6 - \kappa (L_3 \otimes I_2) + \eta (I_3 \otimes J), \kappa, \eta \geq 0. \quad (13)$$

$$N = \text{diag}(n_1, n_1, n_2, n_2, n_3, n_3), n_j > 0, A = N^{1/2} R N^{1/2}. \quad (14)$$

$$x_{t+1} = A x_t + u_t, x_0 = e_1 \text{ for impulse experiments, } u_t = 0 (t \geq 0). \quad (15)$$

The path Laplacian in Equation (12) couples neighboring levels. The parameter a controls local retention, κ cross-level transport and η within-level coupling. The gains n_1, n_2 and n_3 specify the neutral operator in Equation (14). They are not the order coordinates φ_n of Section 4. A future constitutive law may relate them, but none is assumed in these simulations. All parameters, states and update steps are dimensionless. The matrix A is an effective perturbation-response operator, not a full model of self-maintenance, metabolism or conscious activity.

In this specialization N has no independent dynamical state. It acts instantaneously through a factorization of A . That choice permits an exact analysis and also limits the evidence obtainable from it. A physical mediator with its own relaxation and memory would require additional states and a separate model comparison. A stable response returning toward zero describes the decay of a deviation around the maintained configuration, not disappearance of that underlying configuration.

6.2 Exact spectrum and stability

$$\lambda_{j,\pm} = n(a - \kappa \ell_j \pm \eta), (\ell_1, \ell_2, \ell_3) = (0, 1, 3). \quad (16)$$

Proposition 2. For a homogeneous neutral gain n , the six eigenvalues are given by Equation (16). **Proof.** The path Laplacian has eigenvalues 0, 1 and 3. The within-level exchange matrix J has eigenvalues $+1$ and -1 . The Kronecker factors commute and admit tensor-product eigenvectors. Homogeneous N equals n times the identity, so conjugation multiplies every eigenvalue of R by n . This yields the six values in Equation (16).

$$\rho(A) = \max_{j,\pm} |n(a - \kappa \ell_j \pm \eta)| < 1. \quad (17)$$

$$\|x_t\|_2 \leq r^t \|x_0\|_2 + \frac{1-r^t}{1-r} \sup_{0 \leq k < t} \|u_k\|_2, r = \rho(A) < 1. \quad (18)$$

A is real symmetric, so its Euclidean operator norm equals its spectral radius. If r is strictly below one, repeated substitution and the geometric-series bound give Equation (18). This is a sufficient input-to-state bound for bounded input in this model. At spectral radius one, unforced symmetric dynamics remain power bounded, but a persistent input can accumulate along a unit mode. Therefore near-unit propagation must

not be treated as a general robustness guarantee. For non-normal empirical operators, spectral radius alone also fails to bound transient amplification.

For the plotted domain $a = 0.82$, $\kappa = 0.12$ and $0 \leq \eta \leq 0.35$, all eigenvalues before multiplication by n are positive and the largest is $a + \eta$. The stability boundary is therefore $n = 1/(a + \eta)$. Outside this domain, Equation (17), rather than that simplified expression, determines stability. No spectral threshold is identified with a threshold of consciousness.

7 Differentiation and propagation beyond scalar stability

$$c = \frac{(1, 1, 1, 1, 1, 1)^T}{\sqrt{6}}, d = \frac{(1, -1, 1, -1, 1, -1)^T}{\sqrt{6}}. \quad (19)$$

$$C_T = \|A^T c\|_2 = |n(a + \eta)|^T, D_T = \|A^T d\|_2 = |n(a - \eta)|^T. \quad (20)$$

The normalized vectors c and d are spatially uniform common and contrast modes. In Equation (20), the superscript T is an integer power denoting the measurement horizon; matrix transposition is written with a separate upright T in other equations. C_T measures common-mode retention and D_T measures retention of the declared difference between the two role coordinates. D_T is not a general measure of semantic differentiation. Its meaning is restricted to this explicitly chosen perturbation direction.

$$n = \frac{r}{a + \eta} \Rightarrow C_T = r^T, D_T = \left| \frac{r(a - \eta)}{a + \eta} \right|^T. \quad (21)$$

Proposition 3. Common-mode persistence does not determine contrast retention. For parameters with nonnegative eigenvalues and the common mode dominant, choose $n = r/(a + \eta)$. Then C_T remains fixed, while D_T varies according to Equation (21). For a positive η below a , the ratio $(a - \eta)/(a + \eta)$ decreases as η increases. Hence greater reciprocal coupling can suppress the declared contrast even when the overall spectral radius is unchanged. This is a conditional mathematical statement about mode structure, not a diagnosis of hypersynchrony.

$$G_T = \sum_{t=0}^T e_6^T A^t e_1. \quad (22)$$

G_T summarizes the response at the self-role coordinate of level 3 to an impulse at the content-role coordinate of level 1. It is a finite-horizon transfer sum, not mutual information or a consciousness index. When $\kappa = 0$, the three levels are disconnected and G_T is exactly zero. The spectral radius, C_T and D_T can nevertheless equal those of a connected system. At least three separate questions therefore remain: whether a response persists, whether a selected difference survives and whether information can influence another declared level.

These distinctions constrain a useful BFG balance interpretation. Maximizing persistence or coupling cannot serve as a universal objective. A functional system may need sufficient retention, preserved discriminability and appropriate propagation, with the acceptable ranges determined by its task and physical realization. The word balance acquires empirical content only when those ranges and the relevant functions are independently specified.

8 Reproducible numerical illustrations

8.1 Parameters and computational checks

The reference construction uses $a = 0.82$, $\kappa = 0.12$, $\eta = 0.10$ and gains $(1, 1, 1)$. Impulse trajectories cover 60 update steps; reported summary metrics use $T = 20$. The exact spectrum was checked against numerical

diagonalization for 1,485 homogeneous cases: 11 values of κ from 0 to 0.25, 15 values of η from 0 to 0.35 and 9 values of n from 0.5 to 1.3. The largest absolute eigenvalue discrepancy was 9.99×10^{-16} . This verifies the implemented formulas within floating-point precision; it does not test physiology.

Six cases are compared in Table 3. Weak mediation changes all gains to 0.65. Disconnected levels sets κ to zero. Contrast suppression raises η to 0.32 and sets the homogeneous gain to 0.92/1.14 so that the spectral radius remains 0.92. The middle-interface case changes only the second gain to 0.35. The instability case raises all gains to 1.12. These named conditions describe mathematical interventions, not clinical categories.

Table 3. Exact-model numerical summaries at horizon 20. Retention and transfer are dimensionless. Case names have no diagnostic meaning.

Case	$\rho(A)$	C_{20}	D_{20}	G_{20}
Reference	0.9200	0.189	0.0014	0.5727
Weak mediation	0.5980	3.42e-05	2.54e-07	0.0125
Disconnected levels	0.9200	0.189	0.0014	0.0000
Contrast suppression	0.9200	0.189	1.31e-08	0.5438
Middle interface reduced	0.8174	0.016	6.97e-05	0.0613
Linear instability	1.0304	1.82	0.0135	2.6301

8.2 Stability surface and retention tradeoff

Linear stability surface at $a = 0.82$ and $\kappa = 0.12$

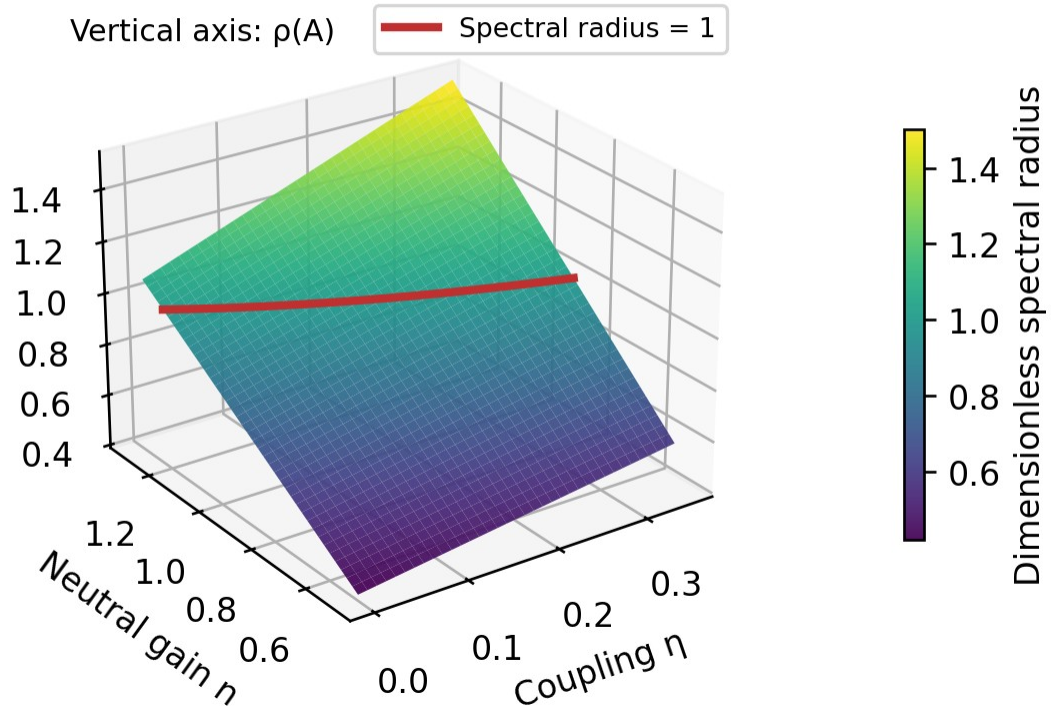


Figure 3. Three-dimensional spectral surface for the homogeneous model. The red curve marks spectral radius one. The surface depicts an exact linear stability boundary; it is not a landscape of consciousness or clinical severity.

Figure 3 displays the sensitivity of stability to neutral gain and within-level coupling. At $\eta = 0.10$, the boundary is $n \approx 1.086957$. The reference gain lies below it, whereas $n = 1.12$ produces spectral radius 1.0304. A finite trajectory can remain moderate initially even in this unstable case, so a short observation window cannot substitute for an asymptotic calculation.

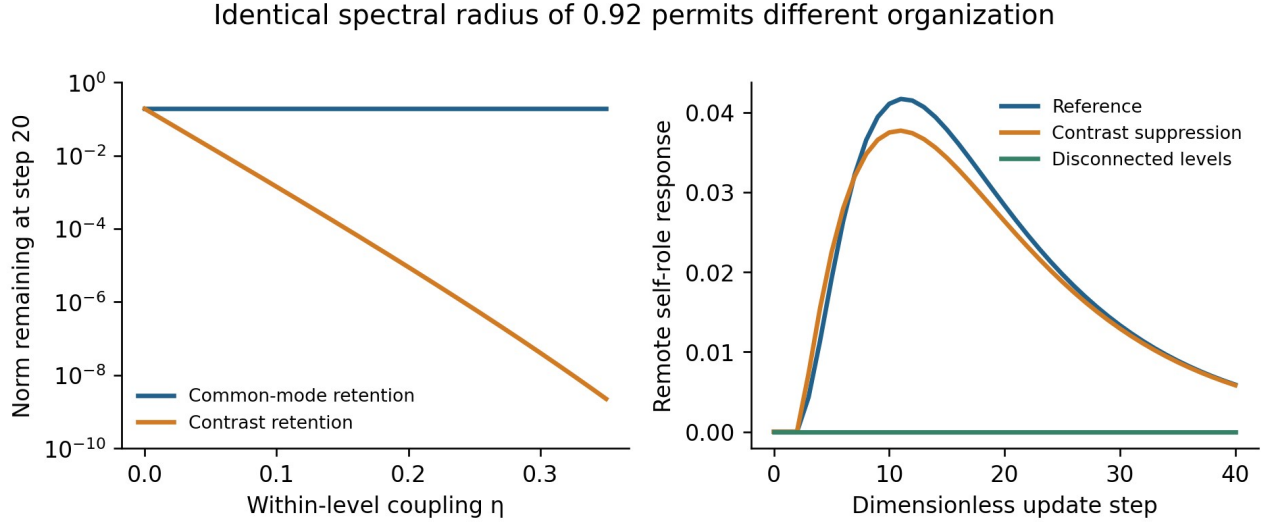


Figure 4. Left: increasing within-level coupling at a fixed spectral radius of 0.92 leaves common-mode retention unchanged while reducing contrast retention. Right: remote impulse responses distinguish connected, contrast-suppressed and disconnected constructions. All curves are model-generated.

The reference and contrast-suppression cases both retain approximately 0.188693 of the common mode at step 20. Their contrast retentions are approximately 1.40168×10^{-3} and 1.30936×10^{-8} , a ratio of about 107,051. Meanwhile, disconnecting levels leaves both uniform-mode retentions unchanged but removes remote transfer entirely. Figure 4 makes these two failures of a scalar stability description visible. These results motivate separate empirical measurements; they do not select universal biological thresholds.

8.3 Perturbation trajectories

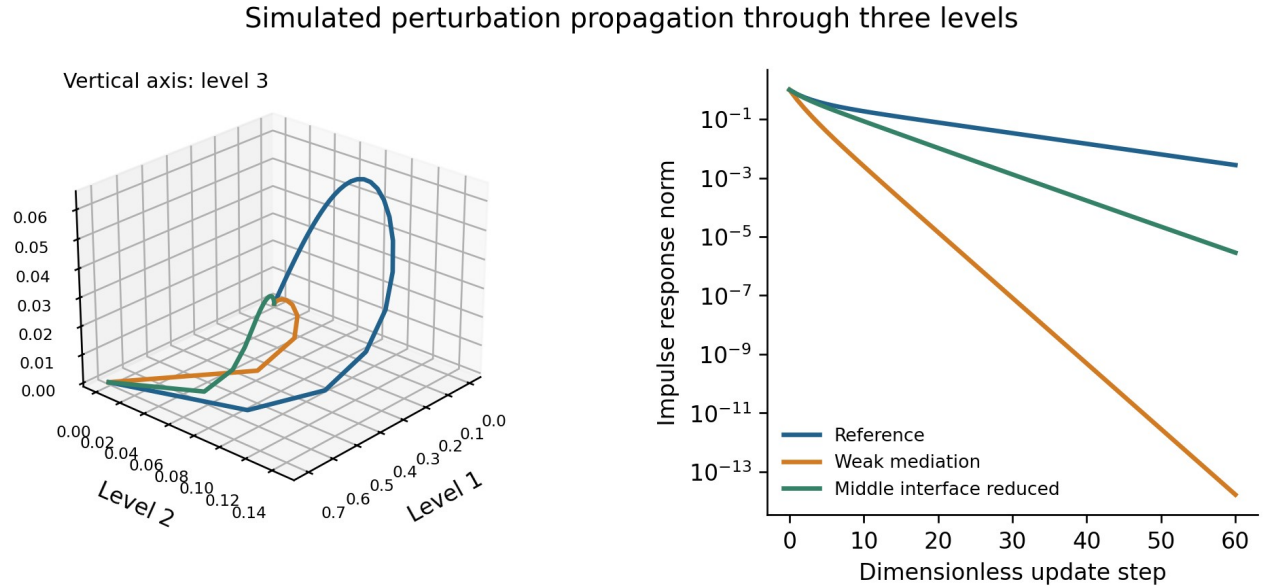


Figure 5. Left: three-dimensional trajectories of the common-mode amplitude at each level after a localized impulse. Right: corresponding response norms. Both axes and time are dimensionless. Trajectories describe deviations around a maintained configuration, with no empirical EEG data.

Reducing the middle gain changes propagation and decay together. Its finite-horizon remote transfer falls from 0.572734 to 0.061295, an approximately 89.3% reduction. That difference is a property of the stipulated gain intervention. A biological manipulation could affect several parameters at once, and the same signal change could have multiple causes. The figure therefore illustrates a prediction conditional on an intervention model rather than a unique neural signature of mediation.

9 Identifiability and the role of the mediator

9.1 An exact direct coupling control

$$A_{direct} = M R M \Rightarrow y_t^{direct} = y_t^{factorized} \text{ for all matched inputs. } (23)$$

Proposition 4. Baseline trajectories do not distinguish this neutral factorization from a direct model with the same effective matrix. Proof. Both models apply the same A to the same state and add the same input. Equality at the initial step implies equality at every subsequent step by induction. Applying the same observation map preserves equality. This is an exact equivalence, not a small-sample limitation. The computed maximum difference was zero across 128 matched, randomly driven updates with seed 20261001.

$$\tilde{R} = \tilde{M}^{-1} A \tilde{M}^{-1} \Rightarrow A = \tilde{M} \tilde{R} \tilde{M}, \tilde{M} > 0. (24)$$

Equation (24) shows a stronger factorization ambiguity. For any invertible positive diagonal matrix $\tilde{M}$, a corresponding $\tilde{R}$ reproduces A exactly. If $\tilde{R}$ belongs to the allowed model class, baseline observations cannot choose the mediator. Restrictions on R may reduce this ambiguity, but they must be justified independently of the desired conclusion. This extends the representation-equivalence discipline emphasized in [6] to the present finite response model.

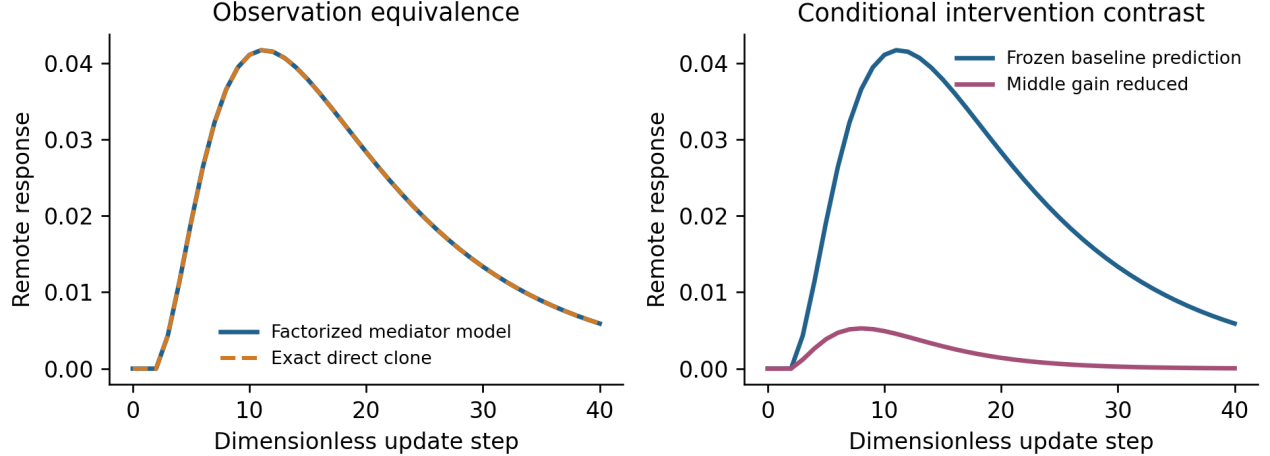


Figure 6. An exact direct-coupling clone overlaps the factorized model in the left panel. The right panel compares a reduced middle gain with a frozen baseline model. The latter difference supports a causal claim only if independent evidence justifies what the intervention changes and what remains fixed.

9.2 What a discriminating intervention would require

$$A^{(q)} = M^{(q)} R M^{(q)}, R \text{ fixed across independently calibrated conditions } q. (25)$$

A mediator interpretation gains testable content when its proposed physical realization can be measured or manipulated and a justified constraint links conditions, as in Equation (25). This is stronger than deleting a fitted term after inspecting an outcome. A valid design needs evidence that the intervention changes the targeted mediator property while retaining the relevant source dynamics, inputs and observation map

sufficiently well. An alternative model must receive the same intervention information and a comparable parameter budget.

The direct competitor can also be allowed a condition-dependent matrix. If that model reproduces all permitted interventions, the experiment does not uniquely identify the factorization. Likewise, algebraic elimination of a genuinely dynamic mediator can create a history-dependent effective interaction. Comparing the full system only with a memoryless rival would then be unfair. A successful BFG test must state whether it identifies a useful effective description, a specific causal carrier or a stronger irreducibility claim; these are different achievements.

The useful outcome of the present control is precise. The model demonstrates how a chosen mediator parameter regulates response, while also showing why that demonstration does not establish that the mediator is physically necessary. Exact clones should be treated as equivalent representations, not opponents that a correct model must outperform.

10 A prospective EEG research program

10.1 Separate empirical targets

The first target is context-sensitive content and self-attribution in participants able to perform tasks and report experience. A feasible research design compares self-generated actions with externally replayed consequences and independently manipulates sensory context. It measures accuracy, agency attribution, response timing and neural responses. Motor preparation, attention, expectation and sensory intensity must be modelled as possible explanations. This is a proposed study, not an experiment conducted for this paper.

The second target is neural organization across sleep, anesthesia and disorders of consciousness. Perturbational studies demonstrate the value of examining the spread and differentiation of induced activity [12–14]. Large-scale EEG work supports multivariate assessment [15], and cognitive motor dissociation shows why motor output and task-related neural responses must be considered separately [16]. A BFG study should compare its declared roles against established measurements and clinical assessments rather than replacing them with a new label.

Psychiatric application requires its own protocol. EEG research has associated impaired Gestalt perception in schizophrenia with altered long-range coordination [17], and a speech study found altered frontal–temporal coupling during talking [18]. These findings motivate questions about context and self-attribution. They do not establish BFG, and they do not show that schizophrenia is a single loss of closure. Psychodynamic concepts such as conflict, defense or attachment require independently defined interview or task measures; no EEG trace is assigned those meanings by assumption.

10.2 Proposed hypotheses

Table 4. Prospective hypotheses with distinct endpoints and explicit alternatives.

Hypothesis	Predicted contrast	What would weaken it
H1 Level-resolved order	Cross-level terms improve held-out response prediction beyond global summaries	No reproducible gain over matched summaries
H2 Differentiated persistence	Contrast and propagation add value beyond response decay alone	Effects disappear after power, arousal and task controls
H3 Identified mediation	A calibrated intervention changes specified return or transport parameters	Equally informed direct or memory models explain the effect
H4 Self and output separation	Self-related task evidence can be dissociated from motor report	The proposed self variable merely follows report demands

H1 and H2 are organizational hypotheses shared in part with other theories. Their success would support the usefulness of the variables, not unique BFG ontology. H3 is more demanding because it tests a separately identified mechanism. H4 tests the proposed role separation while retaining the limits of behavioral and neural evidence. Any association with subjective experience should be stated at the level supported by the available reports and independent assessments.

10.3 Estimation and validation

$$\Delta L = \sum_{i \in D_{test}} \left[\log p_{BFG}(y_i | u_i) - \log p_{rival}(y_i | u_i) \right]. \quad (26)$$

Equation (26) gives one possible primary comparison: held-out log predictive density of neural responses under matched input conditions. It should be accompanied by calibration, residual checks and task-relevant outcomes. Model fitting, normalization, feature selection and threshold selection belong inside training folds. Participant-level and site-level separation are necessary where repeated measurements could leak information. Nested validation addresses the optimism caused by reusing cross-validation for model selection and performance estimation [23].

Before data collection, the protocol should fix the state representation, level definitions, source reconstruction, observation model, input encoding, comparison models and primary time window. Sample size should be chosen through a separate power or precision analysis using realistic noise and missingness, not the deterministic six-state simulation. Uncertainty should reflect participants and sites rather than treating every EEG sample as independent. A subsequent external dataset tests transfer with frozen transformations and parameters or with an explicitly declared adaptation rule.

Strong rivals include a direct recurrent model, an effective model with matched memory, a common-input explanation and an appropriate generative connectivity model. Remove or shuffle a BFG role only under a declared operation that preserves the intended nuisance properties. A shuffled signal that also destroys its spectrum, variance or task structure is not a selective mediation test. Failure results should be retained, including failures of source identifiability, out-of-sample transfer and added predictive value.

TMS and clinical studies require the usual specialist oversight, participant protections and independent ethics review. The present theoretical paper provides no treatment recommendation, diagnostic decision rule or patient-level score. A negative task response is not treated as proof of absent experience.

11 Relation to other accounts and theoretical implications

The BFG proposal overlaps with established research in several ways. Dynamic causal modelling formalizes hypotheses about interacting hidden states and their measured consequences [9,10]. Global neuronal workspace modelling links conscious access to recurrent, broadly available activity [21]. IIT specifies a different formal relation between intrinsic cause–effect structure and phenomenal properties [22]. Perturbational complexity provides an experimentally developed measure rather than an ontology by itself [13,14]. These approaches are meaningful comparators, not interchangeable names for BFG roles.

Table 5. Comparators and the additional work a BFG interpretation would need.

Comparator	Relevant established contribution	BFG comparison requirement
Dynamic causal modelling [9,10]	Explicit generative dynamics and parameter inference	Explain what neutral role constraints add
Workspace model [21]	Recurrent access and distributed availability	Separate self-related organization from accessibility

Comparator	Relevant established contribution	BFG comparison requirement
IIT 4.0 [22]	Formal intrinsic causal structure and phenomenal postulates	Avoid identifying BFG balance with IIT quantities
PCI research [13,14]	Perturbation-based empirical stratification	Demonstrate added value beyond response complexity

The mathematical results sharpen the meaning of balance. Overall persistence can coexist with a loss of the declared contrast or a failure of communication across levels. Stable organization is therefore too broad a category to define healthy function. A clinical interpretation needs preserved capacities, appropriate responsiveness and task-specific evidence of impairment. This also prevents the opposite error of equating every state transition, fluctuation or local imbalance with pathology.

For Fuchs's account [7], a constructive BFG contribution would identify mechanisms that maintain context and ownership relations across a sequence of perceptions or actions, then show how a specified disturbance alters both measured responses and the relevant experience or behavior. The theory would become explanatory through that connection. Temporal continuity remains relevant; additional spatial, organizational and causal structure would explain how it is realized. The present paper supplies an architecture for this work, not a demonstrated solution to the ontology of intentionality.

A useful reconciliation of the BFG sources follows. The direction-error interpretation [1] can motivate a relational ontology. The conditions-of-possibility argument [2] can state the explanatory roles to be investigated. The multiscale and neural manuscripts [3,4] can specify candidate realizations. The witness distinction [5] prevents a measurement from being mistaken for the experience, while the latest reclosure discipline [6] requires independent causal identification. These layers support one research program while preserving different standards of justification.

12 Limitations and conclusion

The main limitations are substantive. The neural role assignments are hypotheses. The observation matrix is not calibrated to EEG. The toy model is linear, reciprocal and memoryless at the neural interface; it contains no metabolic maintenance, learning, anatomical delays or explicit phenomenal content. Its three levels are stipulated, and the order-state example is separate from the response simulation. No constitutive law derives its gains from the original source balance ratio. The mathematical checks demonstrate internal consistency of this finite specialization, not universal BFG validity.

Neither a spectral margin nor a response-mode contrast is sufficient evidence of consciousness. The same effective dynamics admit different neural factorizations, and a direct clone matches all baseline outputs. A physical or ontological conclusion therefore requires evidence not supplied by the numerical example. Likewise, results from disorders of consciousness cannot be transferred directly to psychiatric conditions or psychodynamic constructs without new outcome definitions and studies.

The synthesis nevertheless produces a concrete research advance: it makes differentiated persistence, cross-level propagation and mediator identification separate questions. These distinctions lead to exact results, reproducible figures and prospective comparisons that can fail. The proposed BFG contribution is a structured account of how a candidate functional unity could preserve content and self-related organization across levels. Establishing that such organization is realized in the brain, and determining its relation to subjective experience, remain the empirical and philosophical tasks that follow.

13 Reproducibility and declarations

All reported numerical results and Figures 1–6 are generated by the accompanying Python script `simulate.py`. The package includes parameter definitions, per-case trajectories, machine-readable results,

figure files, bibliography metadata and source-file hashes. No patient data were used, no statistical confidence interval is claimed for deterministic model outputs, and the random seed affects only the driven clone verification. Results were generated with Python 3.12.14, NumPy 2.3.5 and Matplotlib 3.10.8.

Source provenance. References [1–6] are supplied BFG manuscripts, not independent confirmations. Their filenames and hashes are recorded in the supplement. Contributions of Alfredo Pereira Jr. to the joint source [2], and the source corpus’s attribution of Level-0 and conjugate-closure work, are retained by citation. No new coauthorship or endorsement of this synthesis is implied.

Preparation. AI assistance was used for literature retrieval, drafting, implementation and document production. The numerical checks and their limits are reported explicitly. The manuscript is supplied as a preprint draft for the named author’s review. Funding and competing-interest declarations should be completed by the author before journal submission; none is invented here.

Appendix A Mathematical details

A1 Mode decomposition

Let v_j be an orthonormal eigenbasis of L_3 and let $q_{\pm} = (1, \pm 1)/\sqrt{2}$. Then $v_j \otimes q_{\pm}$ form an orthonormal basis of the six-dimensional state space. Substitution into R yields $a - \kappa \ell_j \pm \eta$. Homogeneous conjugation multiplies these values by n . The common and contrast modes used in Section 7 take $v_1 = (1, 1, 1)/\sqrt{3}$. For heterogeneous gains, this simple tensor decomposition no longer diagonalizes A , but A remains symmetric and can be diagonalized directly.

A2 Bounded input and the unit boundary

For $x_t = A^t x_0 + \sum_{k=0}^{t-1} A^{t-1-k} u_k$, symmetry gives $\|A^j\|_2 = r^j$. The triangle inequality and the finite geometric series give Equation (18). If $A c = c$ and every input equals c , then a trajectory starting at zero satisfies $x_t = t c$. Thus power-bounded unforced dynamics at $r = 1$ do not imply bounded response to persistent input. With stochastic input, stationarity requires additional assumptions; the deterministic bound is not a stochastic stability theorem.

A3 Scope of the clone result

An invertible change of hidden coordinates together with the corresponding transformed observation map preserves input–output behavior. Equation (24) concerns a related factorization freedom even with fixed state coordinates. It does not say that all biological mechanisms are interchangeable. It says that data and intervention assumptions must rule out the relevant alternatives. Anatomical measurements, selectively targeted perturbations and independently calibrated neutral responses could constrain that equivalence class; unperturbed fitted trajectories alone cannot.

Appendix B Computational specification

The impulse enters coordinate 1; remote response is coordinate 6. G_{20} includes samples 0 through 20. The initial state is included in each CSV trajectory. Running `python simulate.py` regenerates the numerical outputs and PNG/SVG figures. Parameters are stored in `results.json`. The document builder requires `python-docx`, `lxml` and `Pandoc` for native Word equations.

The script checks the full spectrum across 1,485 cases, zero transfer for disconnected levels, the matched-radius contrast, exact driven-clone equality and alternative-factorization residuals. It stops if a required identity fails. Checks use the full spectral formula even outside the simplified positive-eigenvalue domain.

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